

CS405 - Deep Learning

Project Part 1: Problem Definition, Literature Review, and Dataset Analysis

Due: April 13, 2026, 11:59 PM

Submission through LMS (PDF report)

Overview

This project is designed to give you hands-on experience in solving a deep learning problem using modern architectures (e.g., Transformers, Vision Transformers, Deep CNNs, Deep LSTMs/GRUs). The project is divided into two parts.

Part 1 (this assignment) focuses on problem formulation, literature review, and dataset selection. You will define a clear problem, survey at least 5 relevant research papers, and analyze the dataset you plan to use.

Part 2 will involve implementation, experimentation, and final reporting, where you will train and evaluate your models, compare against baselines, and submit working code along with a comprehensive final report. **Project demo and presentations will be graded under the lab component of the project.**

Learning Objectives

By completing Part 1, you will:

- Formulate a well-defined deep learning problem with clear motivation and challenges.
- Conduct a systematic literature review to understand the evolution of approaches (from classical ML to state-of-the-art deep learning).
- Critically compare different methodologies, architectures, and results from recent papers.
- Select and justify an appropriate dataset for your problem.
- Propose a modern architecture and baseline approach for implementation in Part 2.

Project Selection

You are encouraged to select a topic that aligns with your research interests or career goals. If you have a novel idea not listed below, you may propose it for instructor approval by **April 3, 2026**.

Project Ideas

1. Fine-Grained Visual Classification with Vision Transformers

Classify subtle differences within categories (e.g., bird species, car models) using ViT, DeiT, or Swin Transformer. *Example literature: Dosovitskiy et al., “An Image is Worth 16x16 Words” (ICLR 2021).*

2. **Video Action Recognition with Video Transformers**
Recognize human actions in video using TimeSformer, ViViT, or CNN-LSTM hybrids. *Example: Bertasius et al., "TimeSformer" (ICML 2021).*
3. **Abstractive Text Summarization with BART/T5**
Generate concise, fluent summaries of long documents using pre-trained seq2seq Transformers. *Example: Lewis et al., "BART" (ACL 2020).*
4. **Named Entity Recognition with BiLSTM-CRF and BERT Embeddings**
Extract entities from text using contextual embeddings from BERT/RoBERTa with sequence labeling layers. *Example: Devlin et al., "BERT" (NAACL 2019).*
5. **Multimodal Sentiment Analysis with Fusion Transformers**
Combine text, audio, and video to predict sentiment using MulT or M-BERT. *Example: Tsai et al., "Multimodal Transformer" (ACL 2019).*
6. **Time Series Forecasting with Temporal Fusion Transformer**
Forecast electricity demand or stock prices using attention-based models for temporal data. *Example: Lim et al., "Temporal Fusion Transformers" (IJF 2021).*
7. **Image Super-Resolution with SwinIR or ESRGAN**
Reconstruct high-resolution images from low-resolution inputs using Transformer-based or GAN-based architectures. *Example: Liang et al., "SwinIR" (ICCVW 2021).*
8. **Medical Image Segmentation with U-Net and Vision Transformers**
Segment organs or tumors in medical scans using hybrid CNN-Transformer architectures (e.g., TransUNet). *Example: Chen et al., "TransUNet" (2021).*
9. **Zero-Shot Learning with CLIP or ALIGN**
Classify images from unseen classes using vision-language pre-training. *Example: Radford et al., "CLIP" (ICML 2021).*
10. **Music Genre Classification with Transformers**
Classify music tracks into genres using spectrograms and Transformer-based models. *Example: Huang et al., "Music Transformer" (2018).*
11. **Question Answering with Retrieval-Augmented Generation**
Build a QA system that retrieves relevant documents and generates answers using RAG or FiD. *Example: Lewis et al., "RAG" (2020).*
12. **Graph Neural Networks for Molecular Property Prediction**
Predict molecular properties using GNNs or Graph Transformers. *Example: Ying et al., "Graph Transformer" (ICLR 2021).*
13. **Anomaly Detection in Time Series with Transformers**
Detect anomalies in sensor data or system logs using self-attention mechanisms. *Example: Xu et al., "Anomaly Transformer" (ICLR 2022).*
14. **Object Detection with DETR (Detection Transformer)**
Perform end-to-end object detection using Transformer-based architectures. *Example: Carion et al., "DETR" (ECCV 2020).*

15. Style Transfer with Vision Transformers

Apply artistic styles to images using attention-based feature manipulation. *Example: Park & Lee, “Style Transformer” (ECCV 2020).*

16. Speech Recognition with Conformer or Whisper

Transcribe speech to text using modern hybrid CNN-Transformer architectures. *Example: Gulati et al., “Conformer” (ICASSP 2020).*

17. Reinforcement Learning with Transformer-based Policies

Train an agent to play games or control robots using Decision Transformer. *Example: Chen et al., “Decision Transformer” (ICML 2021).*

Group Formation and Project Uniqueness

- **Group Size:** A maximum of **2 persons per project** is allowed. You may work individually if you prefer.
- **Project Uniqueness:** Each group must choose a project different from all other groups. No two groups may work on the same project topic.
- **Registration Process:** To ensure project uniqueness, you must register your chosen project using the following process:
 1. Form your group (1-2 persons).
 2. Select your preferred project from the list above or a project of your choice.
 3. Submit your group details (names, student IDs) and chosen project topic through a shared Google Form by **April 5th, 2026**.
 4. Project topics will be approved on a first-come, first-served basis. If multiple groups request the same project, only the first to register will be approved; subsequent groups must select an alternative topic.
- **Verification:** It is **your responsibility** to verify that your chosen project is unique before submitting your Part 1 report.

Part 1 Requirements

Your submission should be a single PDF report (6-10 pages) containing the following sections. Use a conference-style format (e.g., double-column, 10pt font) for professional practice.

1. Problem Definition (1-2 pages)

- **Motivation:** Why is this problem important? What are its real-world applications?
- **Task Formulation:** Define the problem formally. Is it classification, regression, generation, segmentation, etc.? Specify inputs and outputs.
- **Challenges:** What makes this problem difficult? Consider data scarcity, noise, class imbalance, long-range dependencies, interpretability, etc.
- **Success Criteria:** How will you evaluate success? List quantitative metrics (accuracy, F1, BLEU, etc.) and qualitative criteria.

2. Literature Review (3-4 pages)

Survey **at least 5 research papers** that address your chosen problem or related tasks. Your review must include:

- A summary of each paper: problem addressed, proposed method, key innovations, and reported results.
- A **comparative analysis** that organizes papers into categories:
 - Classical ML approaches (e.g., SVMs, CRFs, ARIMA)
 - Early deep learning methods (e.g., CNNs, vanilla RNNs)
 - Modern architectures (e.g., Transformers, Vision Transformers, attention mechanisms)
- A comparison table summarizing models, architectures, datasets, and key results across papers.
- Identification of gaps or limitations in existing work that your project will address.
- Proper citations using a consistent format (e.g., IEEE, ACM, or BibTeX).

3. Dataset Analysis (1-2 pages)

- **Dataset Selection:** Name the dataset you will use. Provide a citation if available.
- **Description:** Number of samples, classes (if classification), modalities, input dimensions, etc.
- **Exploratory Analysis:** Include statistics, class distribution, sample visualizations (images, text snippets, spectrograms), and any preprocessing steps required.
- **Data Splits:** Specify train/validation/test splits. If the dataset does not have standard splits, describe your strategy (e.g., k-fold, stratified sampling).
- **Augmentation:** Discuss planned data augmentations (if applicable) and their rationale.
- **Challenges:** Note any dataset-specific issues (bias, imbalance, noise, missing data) and how you plan to address them.

4. Proposed Approach and Baseline (1 page)

- **Baseline Model:** Describe a simple baseline you will implement (e.g., logistic regression, simple CNN, LSTM without attention) to establish a lower bound.
- **Proposed Modern Architecture:** Specify the modern architecture you will implement (e.g., ViT, BART, MulT, TFT). Justify why this architecture is suitable for your problem.
- **Expected Contributions:** What do you expect your implementation to achieve? How will it improve upon the baseline or prior work?

5. Timeline and Implementation Plan

- Outline a plan for Part 2 implementation (3-4 weeks). Include milestones: data preprocessing, baseline implementation, model implementation, hyperparameter tuning, and final evaluation.
- List libraries and tools you will use (PyTorch, TensorFlow, HuggingFace, etc.).

6. References

Provide a complete list of all cited papers and resources in a consistent format.

Submission Guidelines

- **Deadline:** April 13, 2026, 11:59 PM
- **Format:** Submit a single PDF file.
- **Additional Materials (optional):** Along with the PDF, submit a **code skeleton** (e.g., a Jupyter notebook or Python script) that demonstrates data loading and basic preprocessing. This is to ensure you have a working environment for Part 2.
- **Late Policy:** Late submissions will incur a 10% penalty per day, up to a maximum of 3 days.

Academic Integrity

All work must be your own. You are encouraged to discuss ideas with peers, but the writing and analysis must be individual. Plagiarism (copying text, figures, or ideas without attribution) will result in a grade of zero. Properly cite all sources.

Good luck! We look forward to seeing your innovative project proposals.